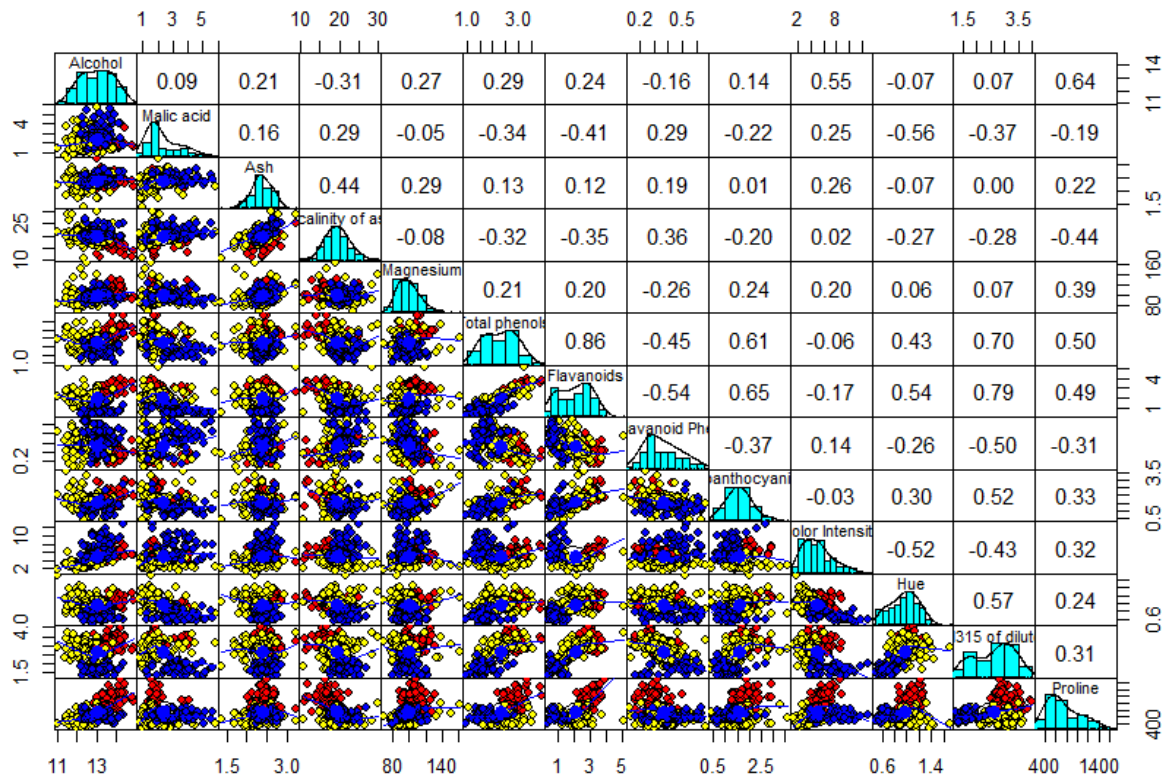
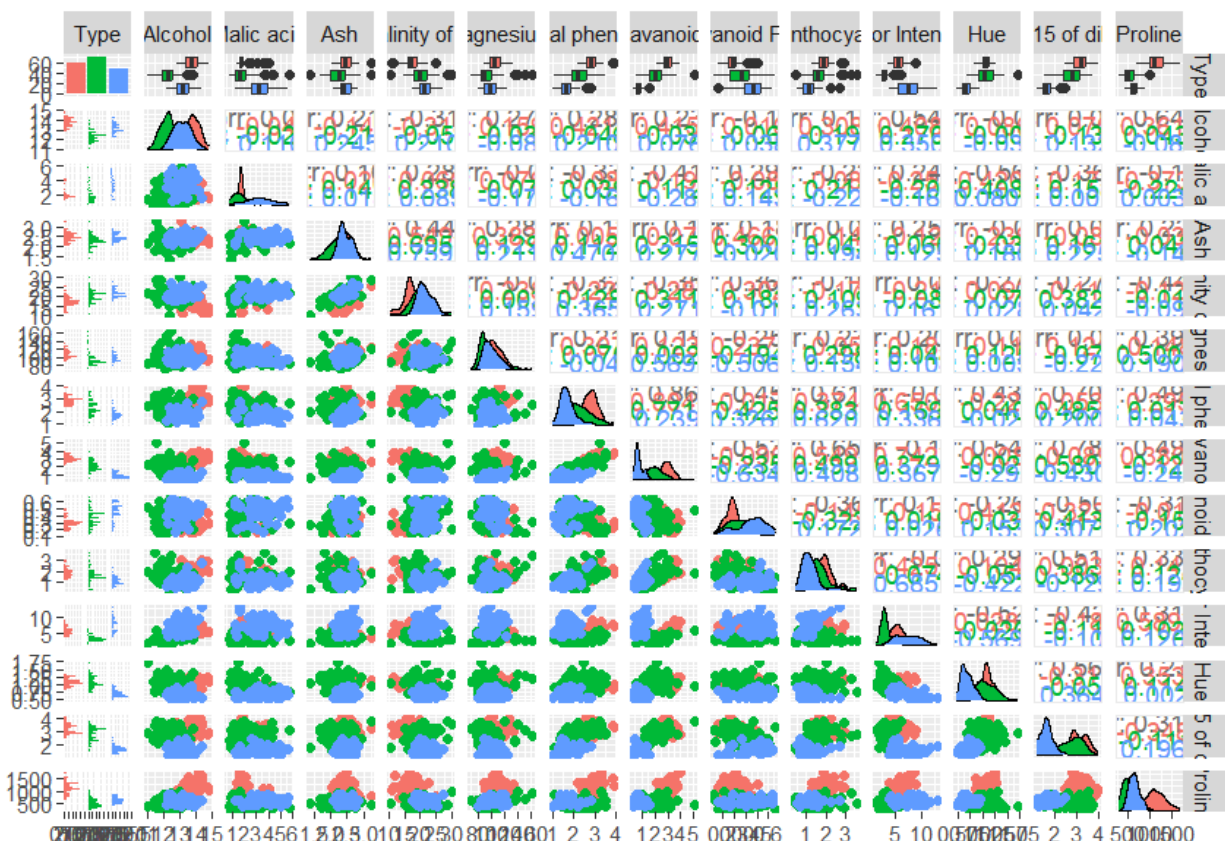


Visualizing the data:





Calculating Principal Components:

```
> summary(principal_components)
```

Importance of components:

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5
Comp.6					
Comp.7					
Standard deviation	2.1631951	1.5757366	1.1991447	0.9559347	0.92110518
0.79878171	0.74022473				
Proportion of Variance	0.3619885	0.1920749	0.1112363	0.0706903	0.06563294
0.04935823	0.04238679				
Cumulative Proportion	0.3619885	0.5540634	0.6652997	0.7359900	0.80162293
0.85098116	0.89336795				
	Comp.8	Comp.9	Comp.10	Comp.11	Comp.12
Comp.13					
Standard deviation	0.58867607	0.53596364	0.49949266	0.47383559	0.40966094
0.320619963					
Proportion of Variance	0.02680749	0.02222153	0.01930019	0.01736836	0.01298233
0.007952149					
Cumulative Proportion	0.92017544	0.94239698	0.96169717	0.97906553	0.99204785
1.000000000					

```
> principal_components$loadings
```



```
Cumulative Var  0.077  0.154  0.231  0.308  0.385  0.462  0.538  0.615  0.692
0.769  0.846
```

```
                Comp.12 Comp.13
SS loadings      1.000   1.000
Proportion Var   0.077   0.077
Cumulative Var   0.923   1.000
```

```
> for (i in 1: 13){
+   print(colnames(wine_matrix)[i])
+   print(principal_components$loadings[i, ])
+ }
```

```
[1] "Alcohol"
      Comp.1      Comp.2      Comp.3      Comp.4      Comp.5      Comp.6      Comp.7
Comp.8      Comp.9      Comp.10      Comp.11
0.14432940  0.48365155  0.20738262  0.01785630  0.26566365  0.21353865  0.05639636
0.39613926  0.50861912  0.21160473  0.22591696
```

```
      Comp.12      Comp.13
0.26628645  0.01496997
```

```
[1] "Malic acid"
```

```
      Comp.1      Comp.2      Comp.3      Comp.4      Comp.5      Comp.6
Comp.7      Comp.8      Comp.9      Comp.10
-0.24518758  0.22493093 -0.08901289 -0.53689028 -0.03521363  0.53681385
-0.42052391  0.06582674 -0.07528304 -0.30907994
```

```
      Comp.11      Comp.12      Comp.13
-0.07648554 -0.12169604  0.02596375
```

```
[1] "Ash"
```

```
      Comp.1      Comp.2      Comp.3      Comp.4      Comp.5      Comp.6
Comp.7      Comp.8      Comp.9
-0.002051061  0.316068814 -0.626223901  0.214175562  0.143025474  0.154474665
0.149170611 -0.170260021 -0.307694445
```

```
      Comp.10      Comp.11      Comp.12      Comp.13
-0.027125389  0.498691415  0.049622373 -0.141218032
```

```
[1] "Alcalinity of ash"
```

```
      Comp.1      Comp.2      Comp.3      Comp.4      Comp.5      Comp.6
Comp.7      Comp.8      Comp.9      Comp.10
-0.23932041 -0.01059050 -0.61208035 -0.06085941 -0.06610294 -0.10082451
0.28696914  0.42797018  0.20044931  0.05279942
```

```
      Comp.11      Comp.12      Comp.13
-0.47931378  0.05574287  0.09168285
```

```
[1] "Magnesium"
```

```
      Comp.1      Comp.2      Comp.3      Comp.4      Comp.5      Comp.6
Comp.7      Comp.8      Comp.9      Comp.10
0.14199204  0.29963400 -0.13075693  0.35179658 -0.72704851  0.03814394
-0.32288330 -0.15636143  0.27140257  0.06787022
```

```
      Comp.11      Comp.12      Comp.13
-0.07128891 -0.06222011  0.05677422
```

```
[1] "Total phenols"
```

```
      Comp.1      Comp.2      Comp.3      Comp.4      Comp.5      Comp.6
Comp.7      Comp.8      Comp.9      Comp.10
```

0.39466085	0.06503951	-0.14617896	-0.19806835	0.14931841	-0.08412230
0.02792498	-0.40593409	0.28603452	-0.32013135		
Comp.11	Comp.12	Comp.13			
-0.30434119	0.30388245	-0.46390791			

[1] "Flavanoids"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9			
0.422934297	-0.003359812	-0.150681900	-0.152294795	0.109025838	-0.018920023
0.060685214	-0.187245357	0.049578492			
Comp.10	Comp.11	Comp.12	Comp.13		
-0.163150510	0.025694093	0.042898829	0.832257065		

[1] "Nonflavanoid Phenols"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9	Comp.10		
-0.29853310	0.02877949	-0.17036816	0.20330102	0.50070298	-0.25859401
-0.59544729	-0.23328465	0.19550132	0.21553507		
Comp.11	Comp.12	Comp.13			
-0.11689586	-0.04235219	0.11403985			

[1] "Proanthocyanins"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9	Comp.10		
0.31342949	0.03930172	-0.14945431	-0.39905653	-0.13685982	-0.53379539
-0.37213935	0.36822675	-0.20914487	0.13418390		
Comp.11	Comp.12	Comp.13			
0.23736257	0.09555303	-0.11691707			

[1] "Color Intensity"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9	Comp.10		
-0.08861670	0.52999567	0.13730621	-0.06592568	0.07643678	-0.41864414
0.22771214	-0.03379692	0.05621752	-0.29077518		
Comp.11	Comp.12	Comp.13			
-0.03183880	-0.60422163	-0.01199280			

[1] "Hue"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9	Comp.10		
0.29671456	-0.27923515	-0.08522192	0.42777141	0.17361452	0.10598274
-0.23207564	0.43662362	0.08582839	-0.52239889		
Comp.11	Comp.12	Comp.13			
0.04821201	-0.25921400	-0.08988884			

[1] "Od280/od315 of diluted wines"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9	Comp.10		
0.37616741	-0.16449619	-0.16600459	-0.18412074	0.10116099	0.26585107
0.04476370	-0.07810789	0.13722690	0.52370587		
Comp.11	Comp.12	Comp.13			
-0.04642330	-0.60095872	-0.15671813			

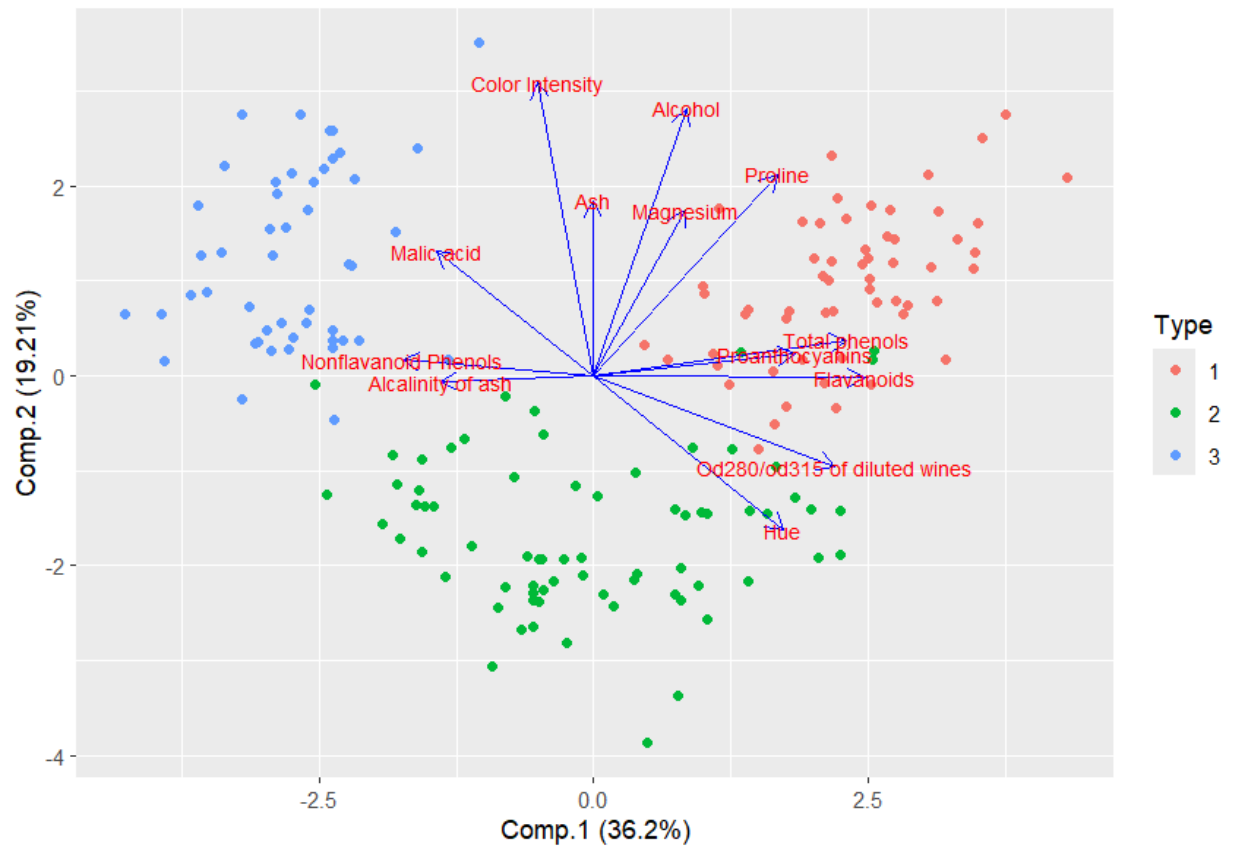
[1] "Proline"

Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Comp.7	Comp.8	Comp.9	Comp.10		

```

0.28675223 0.36490283 0.12674592 0.23207086 0.15786880 0.11972557
-0.07680450 0.12002267 -0.57578611 0.16211600
Comp.11 Comp.12 Comp.13
-0.53926983 0.07940162 0.01444734

```



```
> cm
```

```

      Predicted
Actual  1  2  3
1      13  0  0
2       3 22  0
3       0  0 16

```

```
> accuracy
```

```
[1] 0.9444444
```

```
> data.frame(recall, precision, f1)
```

```

  recall precision  f1
1    1.00    0.8125 0.8965517
2    0.88    1.0000 0.9361702
3    1.00    1.0000 1.0000000

```

Confusion Matrix and Precision/Recall/F1 scores of kNN Model based on PCs

```
> cm1
```

```
      Predicted
Actual  1  2  3
      1 13  0  0
      2  4 20  1
      3  0  0 16
```

```
> accuracy
```

```
[1] 0.9074074
```

```
> data.frame(recall, precision, f1)
```

```
  recall precision      f1
1    1.0 0.7647059 0.8666667
2    0.8 1.0000000 0.8888889
3    1.0 0.9411765 0.9696970
```